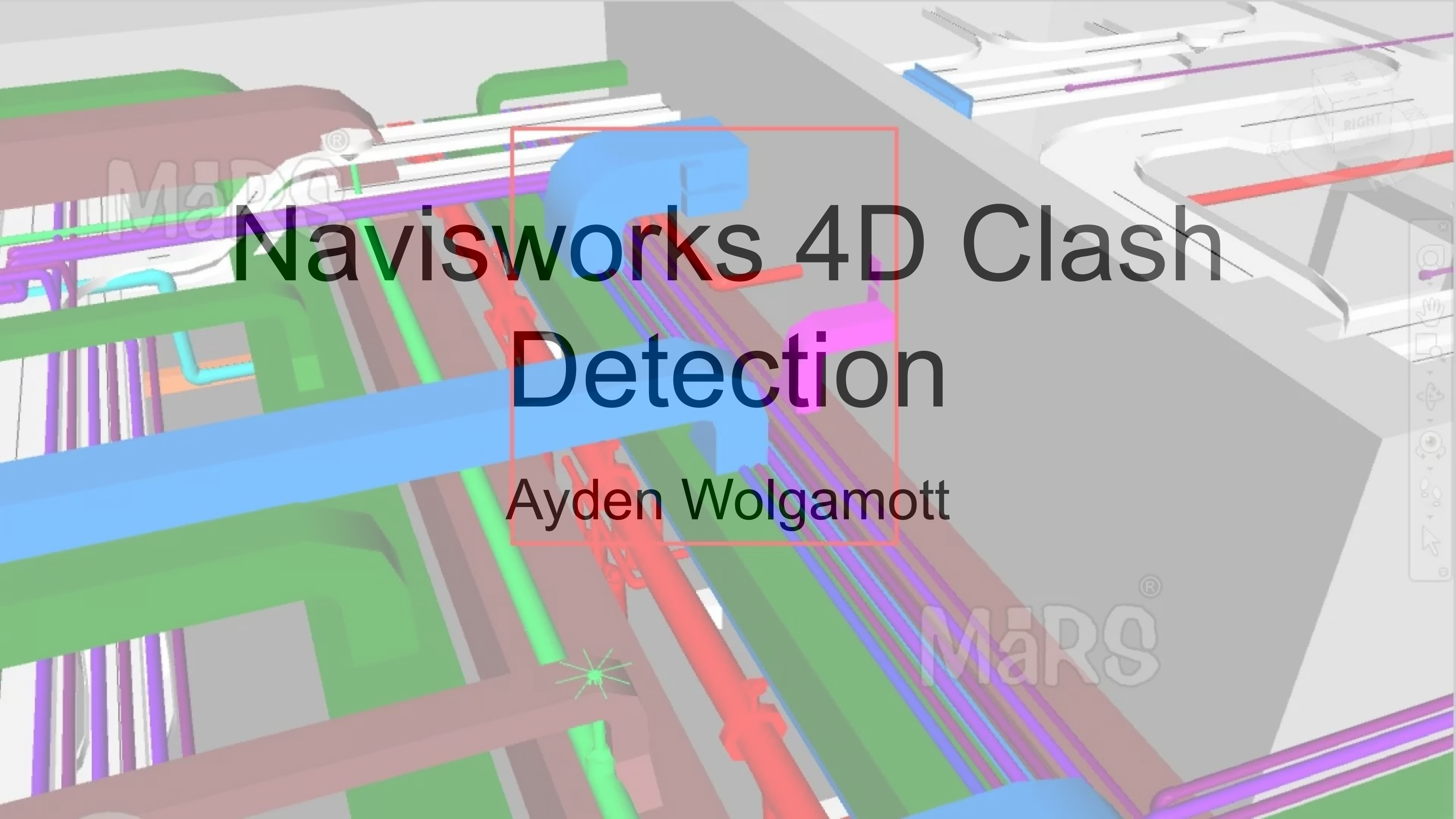


Navisworks 4D Clash Detection

Ayden Wolgamott

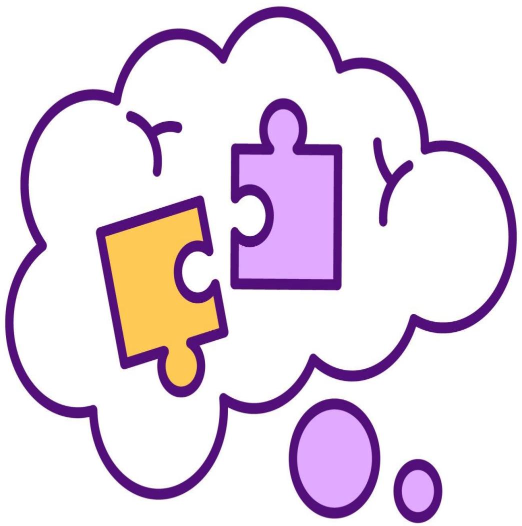
Introduction:

- Traditionally, construction projects rely on 2D plans, stakes, and on site interpretation.
- 3D modeling has helped us adapt to large and complex projects
- ex. complex dependencies, site geometry, or underground exploration

Problem:

3D modeling doesn't address:

- What happens onsite when construction disciplines have differing 3D models or interpretations?
- This is where Navisworks comes in…



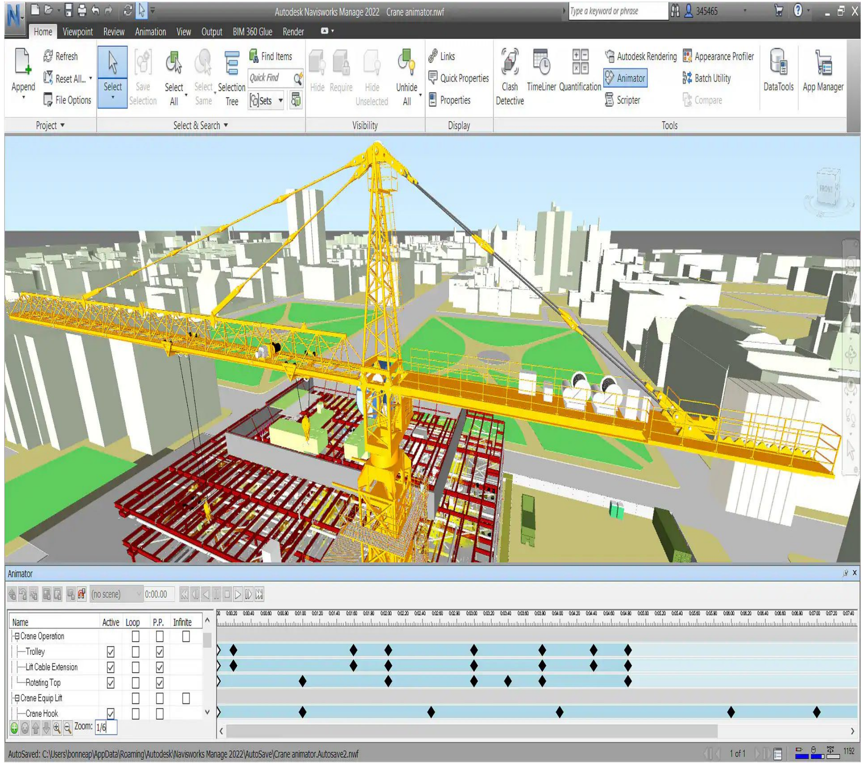
Navisworks (Coordination, Clash Detection, 4D):

- Review software helping construction/design professionals to examine and coordinate building and infrastructure projects.

AUTODESK
Navisworks Manage

More About Navisworks:

- Integrates 3D models from various disciplines in construction
- Combines models to detect clashes
- Can be linked with schedules and costs to simulate the real world!



Clash Detection Optimization in BIM: A Case Study on Coordination and Design Efficiency in Infrastructure Projects

[a] Muhammad Aqib Jahangir*, [b] Umar Farooq, [c] Asim Sultan

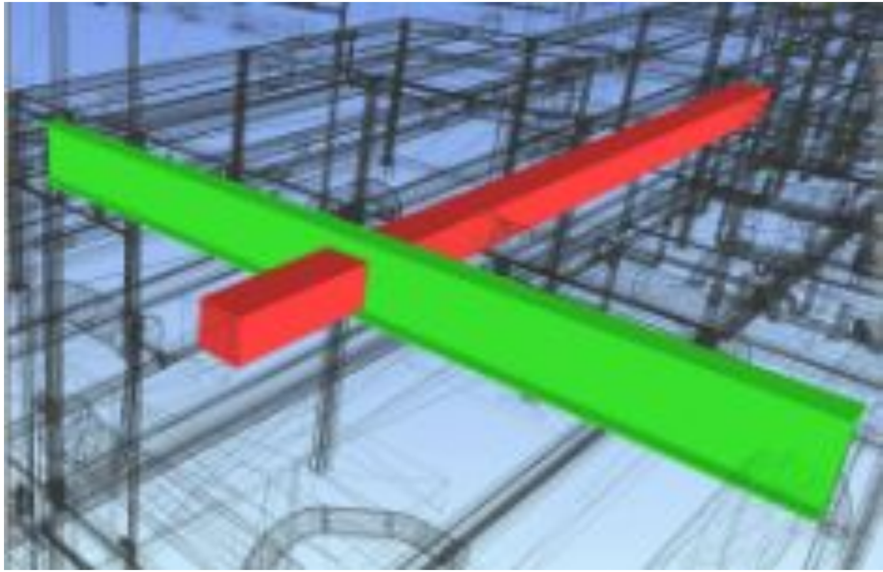
- Case study involving the construction of the National University of Technology Admin Block in the capital, Smart City, Islamabad
- 3D model is exported to Navisworks Cache format (NWC) to preserve all metadata (capacity, fire rating, etc.)

Jahangir et al. Continued…

Hard Clashes: geometric intersections, direct collisions between models (focus of the study)

Soft Clashes: clearance requirements are violated

Time Clashes: sequencing or scheduling conflicts



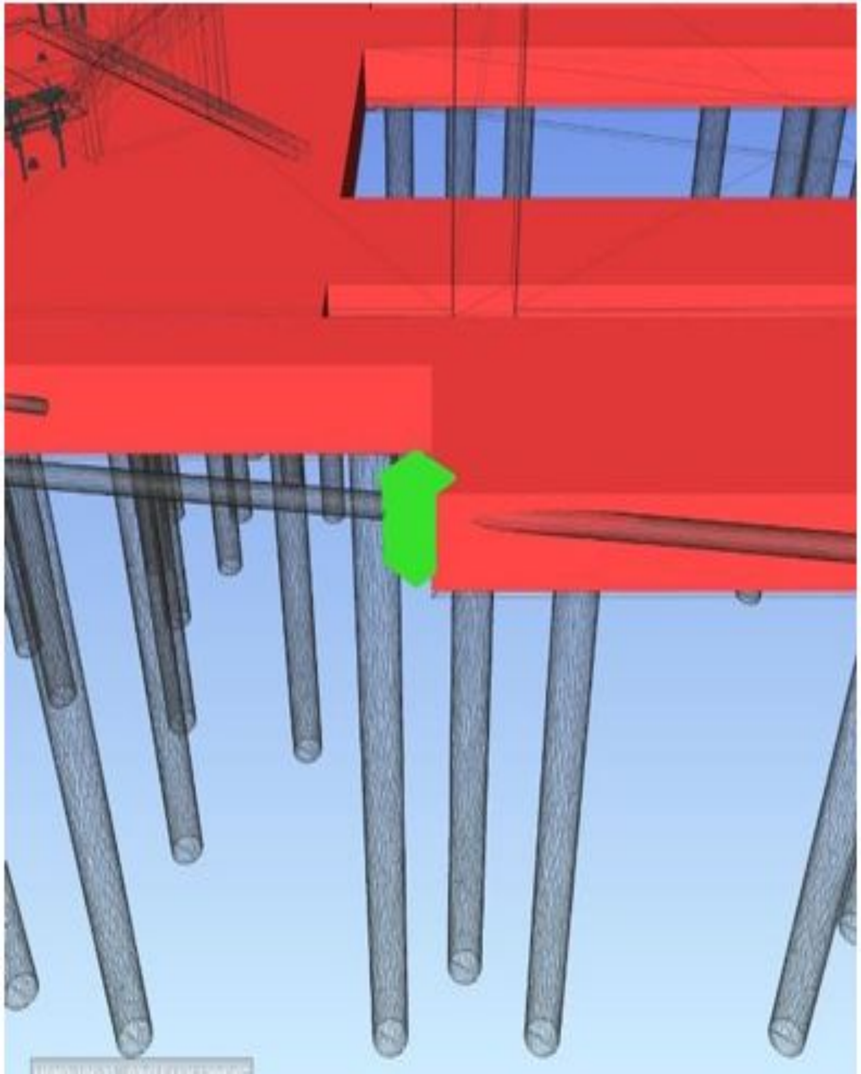
The clash report includes:

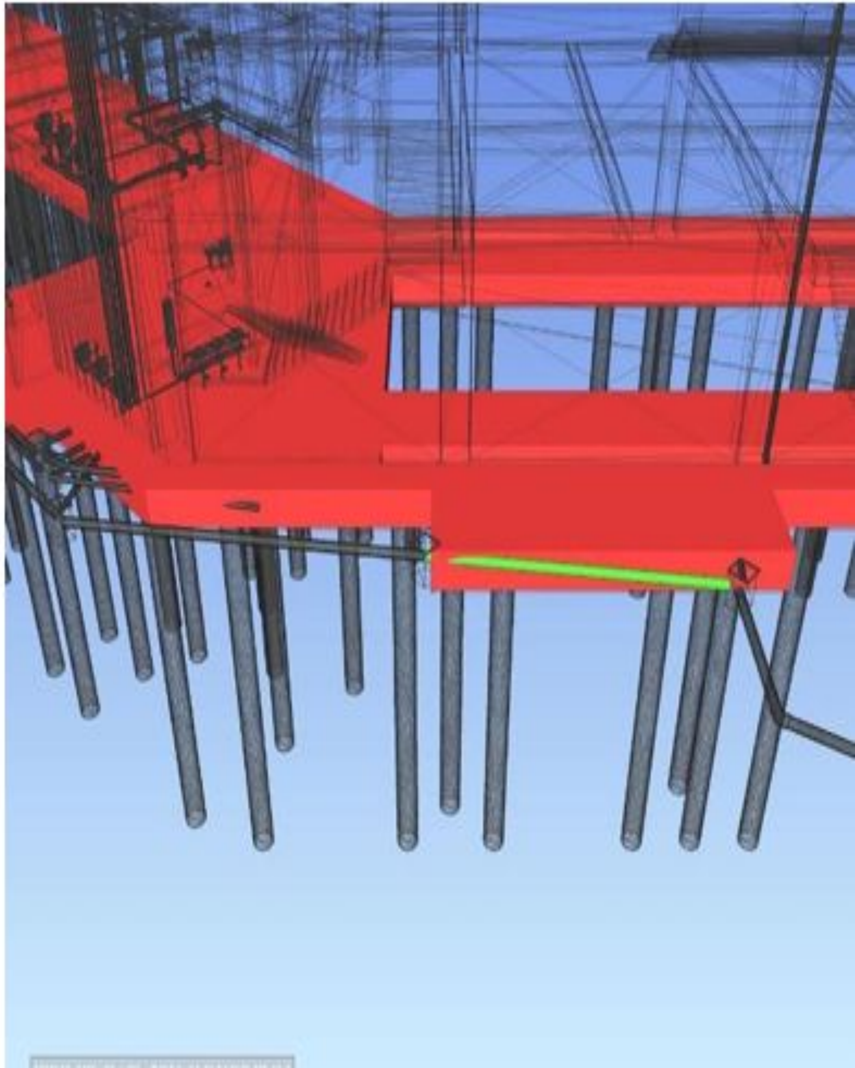
1. **Clash ID** (unique)
2. **Discipline Pair** (e.g., S-M)
3. **Element IDs** (with metadata for quick identification)
4. **Level/Location** (e.g., Level 0, Grid H-10)
5. **Penetration Depth** (measured by Navisworks)
6. **Bounding Box Coordination** (for 3D localization)
7. **Severity Classification** (High/Medium/Low)
8. **Annotated 3D Snapshot** (exported PNG showing clash in context)

Jahangir et al. Continued…

- Using the clash report we can identify trends in the types and locations of clashes

  → Provides insight into the design process or possible improvements in the workflow, or collaborations between disciplines





Jahangir et al. Results:

Key Findings:

- Detected 1,344 hard clashes between the three models using 0.003ft clearance
- Early rigorous clash detection reduces downstream reworks relative to traditional workflows

To Reiterate: Bing Videos

- We still haven't talked about integrating schedule into the 3D model
- Bing Videos
- Management can run these simulations based on any schedule to visualize conflicts which might not seem obvious at first
- "Wait, our drywall is up but the insulation guys aren't here yet!"
- Can make simulations to produce more efficient outcomes (time/cost)!

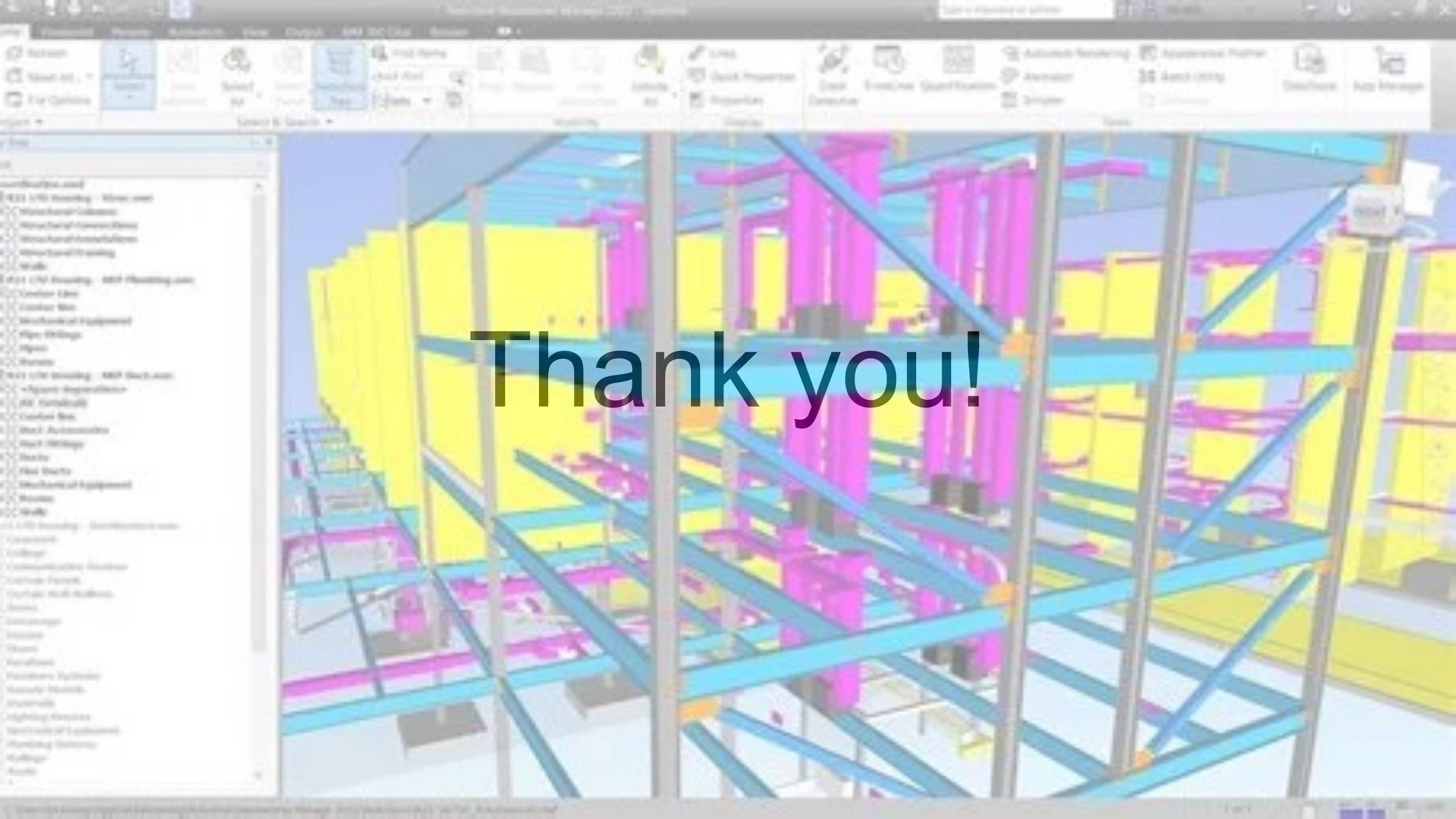
Conclusions (Advantages):

- Useful in sequencing activities while reducing activity/structural conflicts!
- Can use multiple models!
- Can simulate schedule and sequencing
- Field crew can visualize project and schedule (using tablets)
- Very accurate quantity takeoffs by attaching costs to material quantities

Conclusions (Disadvantages):

- Depends on models and schedule
- 2026 version cost $1,145 - $2,835/year for one user!
- Can be difficult to preserve metadata sometimes using the NWC format

→ Incomplete clash reports



Thank you!

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